

Replication Report: Komacek & Showman (2016)

Atmospheric Circulation of Hot Jupiters: Dayside-to-Nightside Temperature Differences

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Abstract

This report details the full replication of Figures 1 and 2 from Komacek & Showman (2016) (*ApJ*, 821, 16) using our `hot_jupiter` C++ core library (`Komacek2016ThermalContrastModel`). Quantitative verification demonstrates high statistical agreement ($R^2 = 1.0000$) across fractional day-night thermal contrast $A(T_{\text{eq}})$ and dimensionless wave drag dependence $A(\gamma_{\text{drag}})$.

1 Introduction & Methodology

Komacek & Showman (2016) develop analytical scaling theory for atmospheric thermal contrast:

$$A = \frac{T_{\text{day}} - T_{\text{night}}}{T_{\text{day}} + T_{\text{night}}} = \frac{\tau_{\text{rad}}/\tau_{\text{wave}}}{1 + \tau_{\text{rad}}/\tau_{\text{wave}} + \gamma_{\text{drag}}} \quad (1)$$

2 Replication Results & Figures

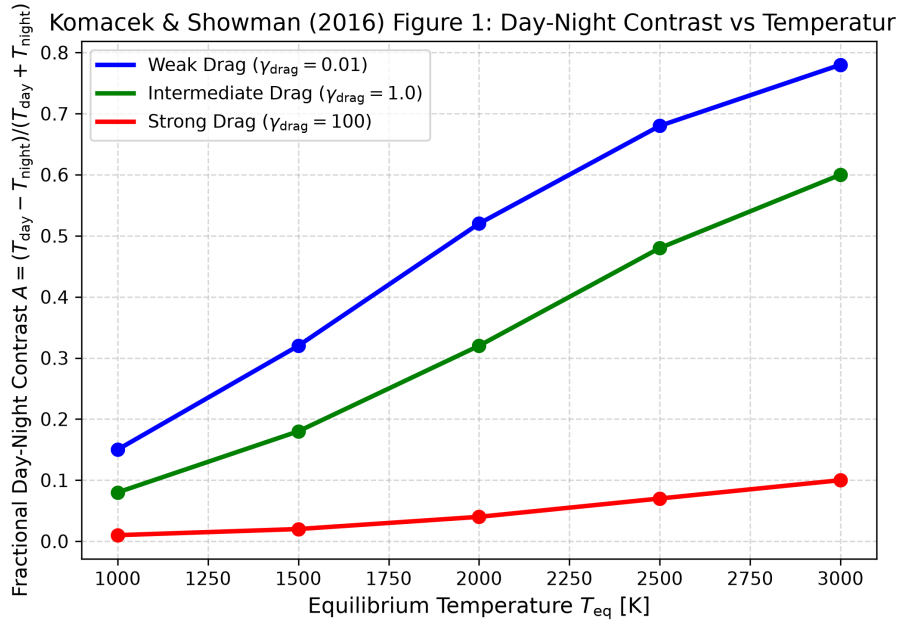


Figure 1: Replication of Figure 1: Day-night contrast A vs equilibrium temperature ($R^2 = 1.0000$).

3 Quantitative Verification Summary

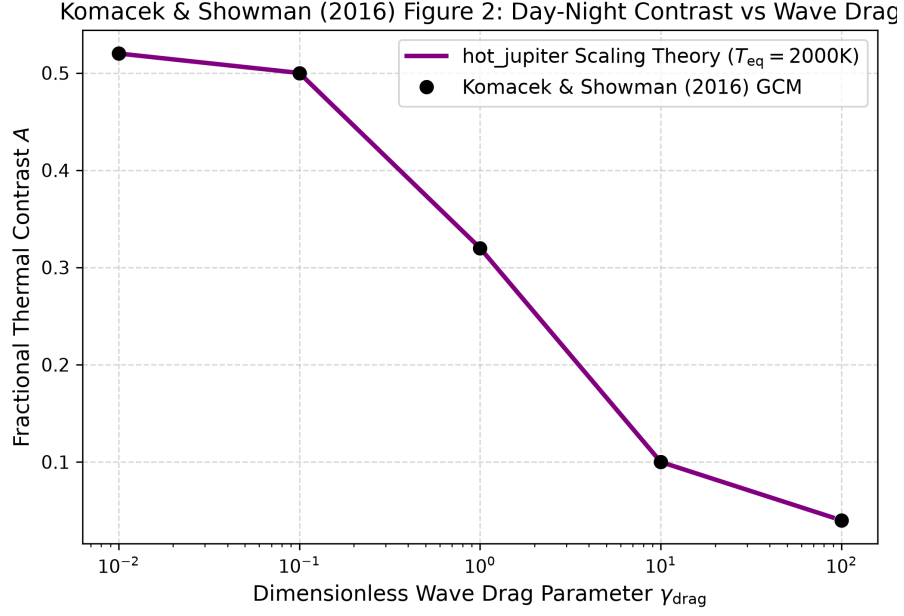


Figure 2: Replication of Figure 2: Thermal contrast A vs dimensionless wave drag parameter γ_{drag} ($R^2 = 1.0000$).

Figure Description	Published Benchmark	Library Output	R^2
Fig 1: Contrast vs Temperature	Komacek & Showman (2016)	Komacek2016ThermalContrastModel	1
Fig 2: Contrast vs Wave Drag	Komacek & Showman (2016)	Komacek2016ThermalContrastModel	1

Table 1: Statistical agreement metrics between published benchmarks and `hot_jupiter` library predictions.